



GSR-Net: enhancing real-time UAV remote sensing object detection via a lightweight transformer model

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Abstract

Remote sensing image detection using deep learning has emerged as a pivotal research area. Recent studies have focused on enhancing feature extraction and multi-scale fusion, often at the cost of increased inference latency. In UAV remote sensing, achieving a balance between detection accuracy and inference speed is crucial. This paper introduces GSR-Net, a lightweight Transformer-based model tailored for real-time UAV remote sensing object detection. GSR-Net integrates HGNetv2 with Ghost convolution and ARConv, specifically designed for small object detection, to create an efficient and lightweight architecture. This design significantly reduces model parameters and computational complexity without sacrificing detection accuracy. Furthermore, GSR-Net adopts a streamlined feature fusion strategy (SFF) that fuses only low-noise upper-level features, optimizing performance. An improved particle swarm optimization algorithm is introduced for hyperparameter tuning during training, shifting computational burden from inference to training. Experiments on the VisDrone2019 dataset demonstrate GSR-Net's superiority, achieving 93.9 FPS on an NVIDIA RTX4090D GPU, outpacing RT-DETR by 35.8 FPS, with improved AP and AP50 by 1.6 and 2.2%, respectively, using 19.6M fewer parameters and only 50.7 GFLOPs. Real-world UAV experiments further validate GSR-Net's robustness against frame loss and missed detections, offering a practical solution for drone inspection applications. The code is open source at <https://github.com/mpwmpw/GSR-Net>.

Keywords UAV imagery · Real-time object detection · GSR-Net · HGNetv2 · Streamlined feature fusion · Particle swarm optimization algorithm

1 Introduction

With the rapid advancement of computational image processing technology [1–3] and the deep integration of aerial remote sensing systems, unmanned aerial vehicles (UAVs) are evolving from traditional flight platforms into intelligent aerial edge-computing systems capable of real-time decision-making. This transformation enables UAVs to play an increasingly vital role in various application scenarios such as urban surveillance [4], traffic management [5, 6], disaster response [7], agricultural inspection [8], and military reconnaissance [9]. However, a key technical challenge remains: achieving low-latency inference while maintaining high detection accuracy in aerial visual systems [10–12].

Compared to satellite imagery, UAV-captured images offer higher resolution, greater flexibility, and lower cost [13], but also pose new difficulties. High-speed flight requires real-time processing of large volumes of image frames. Insufficient inference speed can result in delayed visual feedback, frame drops, and missed detections [14–16]. To enhance

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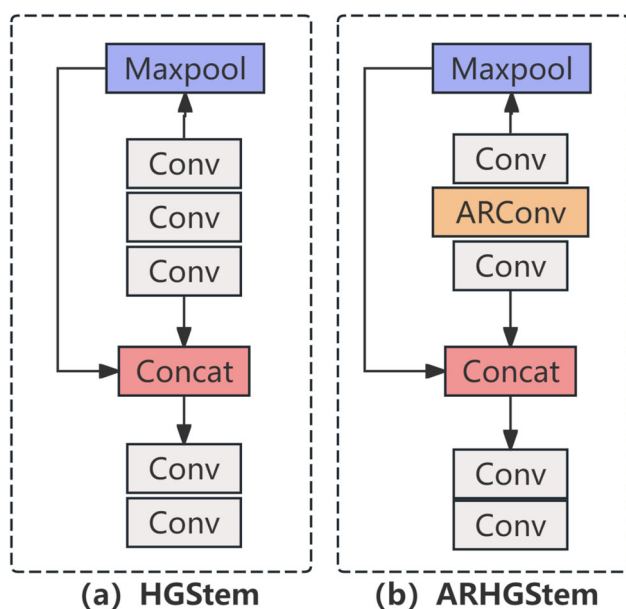


Fig. 1 Comparison of HGStem and GHGStem structures

detection performance, recent studies have introduced optimized multi-scale fusion structures [17–19] and various attention mechanisms [20, 21] to improve feature representation. While effective, these approaches often increase computational complexity, thereby limiting their applicability to real-time UAV scenarios and making them difficult to deploy on embedded platforms [22]. Convolutional Neural Networks (CNNs) have been the foundation of traditional object detection methods. Models such as RetinaNet [23], the YOLO series [24–28], and Faster R-CNN [29] have demonstrated impressive performance on general visual datasets. However, in aerial remote sensing imagery, these CNN-based detectors face challenges such as detecting small objects, handling occlusions, accommodating large scale variations, and maintaining real-time inference [30].

The introduction of Transformers has opened new directions in object detection. Representative models such as DETR [31] and RT-DETR [32] leverage self-attention mechanisms to model global contextual relationships, significantly enhancing object localization and feature association. Nevertheless, similar to CNN-based frameworks, these Transformer detectors are primarily designed and optimized for general-purpose datasets such as COCO, where object scales are relatively uniform, category diversity is high, and viewpoints resemble human perspectives [33, 34]. Influenced by these dataset characteristics, mainstream detection frameworks often adopt a three-stage multi-scale fusion design (P3+P4+P5) to balance speed and accuracy. However, in UAV-based remote sensing imagery, this complex fusion strategy increases computational cost and performs subop-

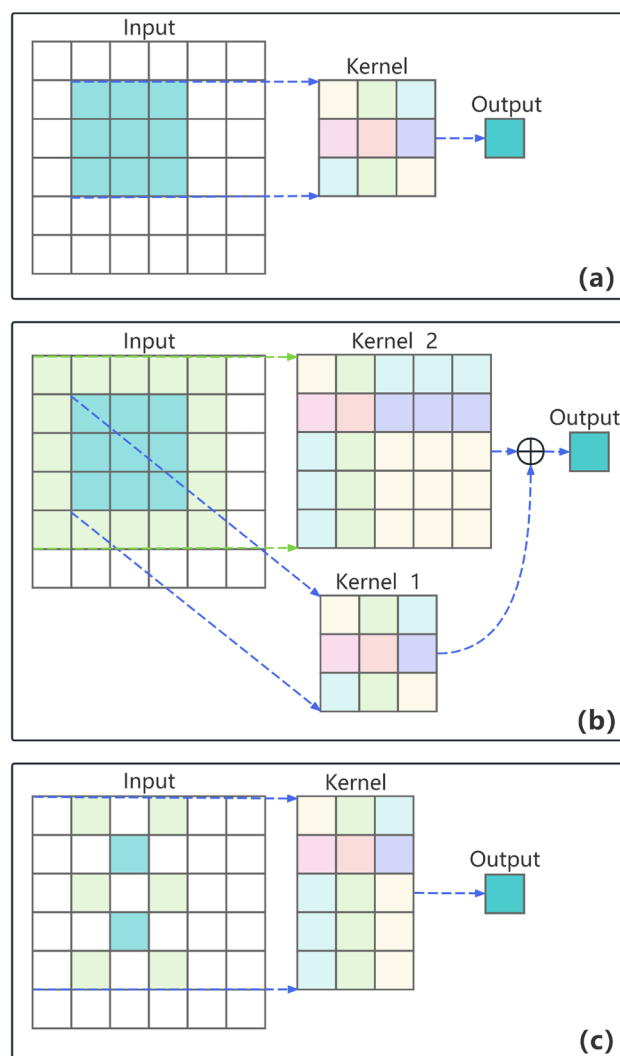
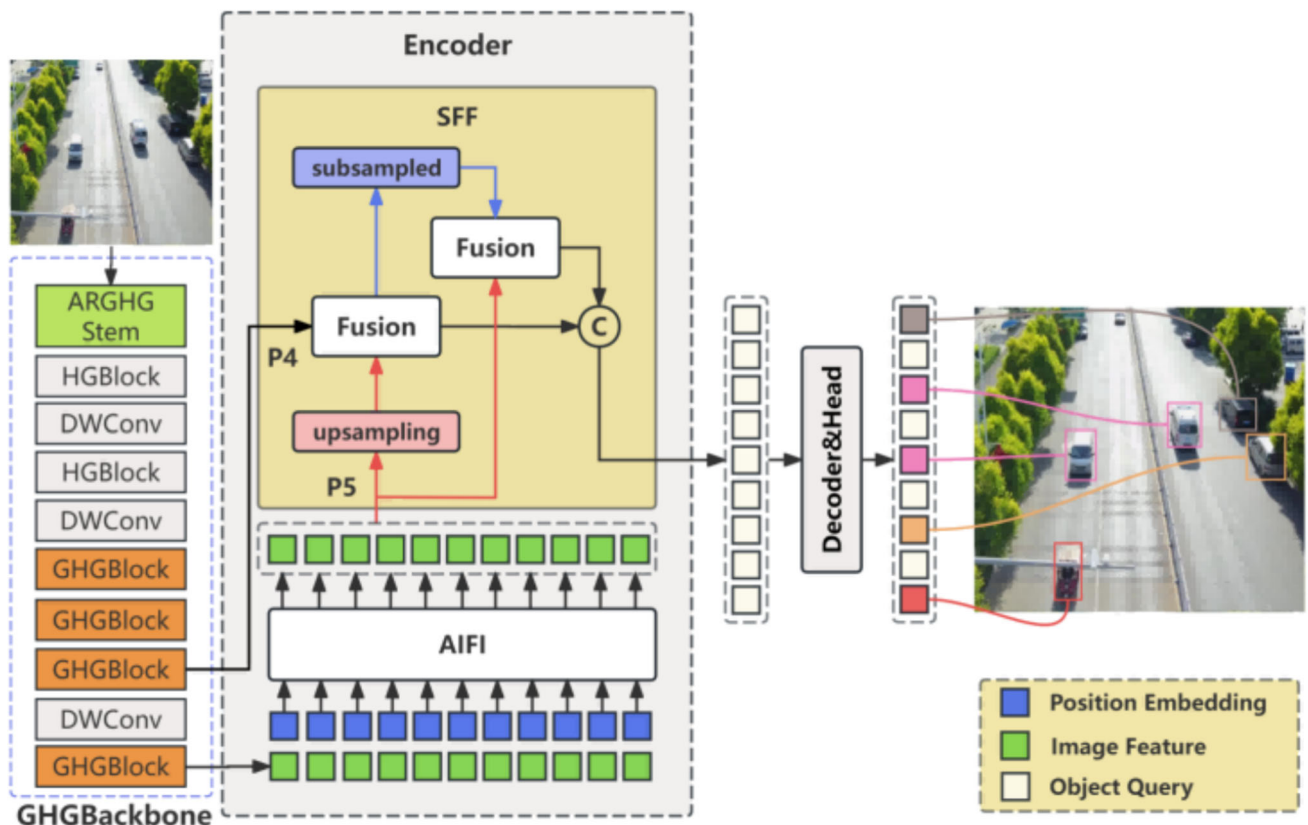
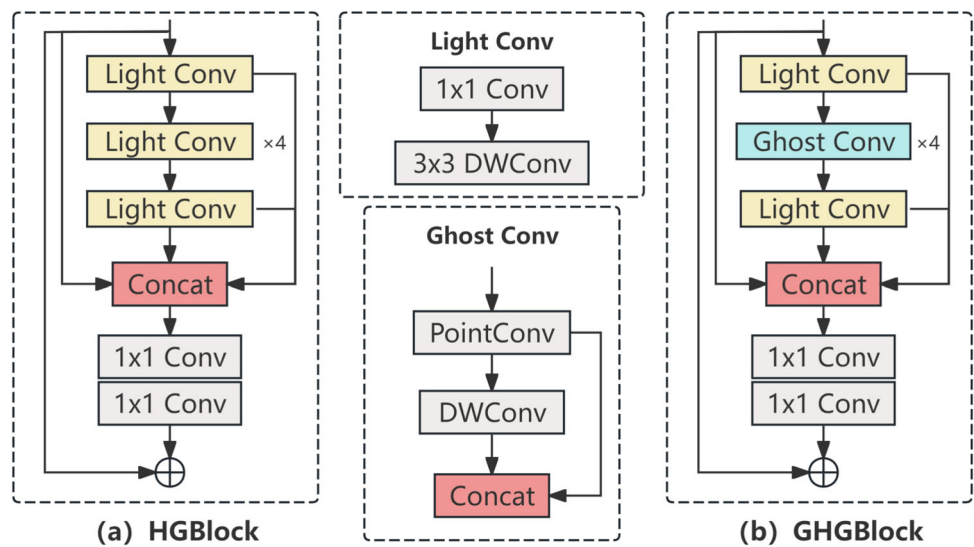


Fig. 2 Comparison of different convolution forms. (a) shows the structure of a standard convolution; (b) illustrates multi-scale convolution; (c) illustrates ARConv used in this paper

timally under the challenges of large object scale variation and sparse object distribution [9, 34].

In terms of backbone design, ResNet-50 has been widely adopted across detection frameworks due to its strong feature extraction capacity and depth control [35]. However, as a structure optimized for general vision tasks, ResNet-50 also exhibits limited adaptability to remote sensing imagery [36]. Aerial images typically contain smaller objects, complex backgrounds, significant scale variations, and abundant visual noise [37]. The repeated downsampling operations in ResNet-50 may lead to excessive spatial compression in shallow layers, causing loss of small object features and thus reduced detection accuracy [38]. More importantly, the large number of parameters and high computational cost of ResNet-50 hinder its deployment on resource-constrained embedded UAV platforms [39]. Therefore, there is a pressing

Fig. 3 Comparison of HGBlock and GHGBlock structures**Fig. 4** GSR-Net model structure diagram

need for a lightweight and task-adaptive backbone architecture that is better suited for small object representation in remote sensing scenes, enabling accurate and efficient real-time detection.

To address these challenges, we propose GSR-Net, a lightweight Transformer-based object detection framework derived from the RT-DETR architecture and specifically tailored for real-time and small object detection in UAV-

based remote sensing imagery. First, we construct an efficient lightweight backbone by integrating HGNetv2 [32] with Ghost convolution, significantly reducing both the model's parameter count and computational overhead while maintaining strong feature extraction capability. Secondly, considering that most objects in remote sensing images are small and the Transformer architecture is naturally good at global modeling, we abandoned the traditional multi-scale fusion

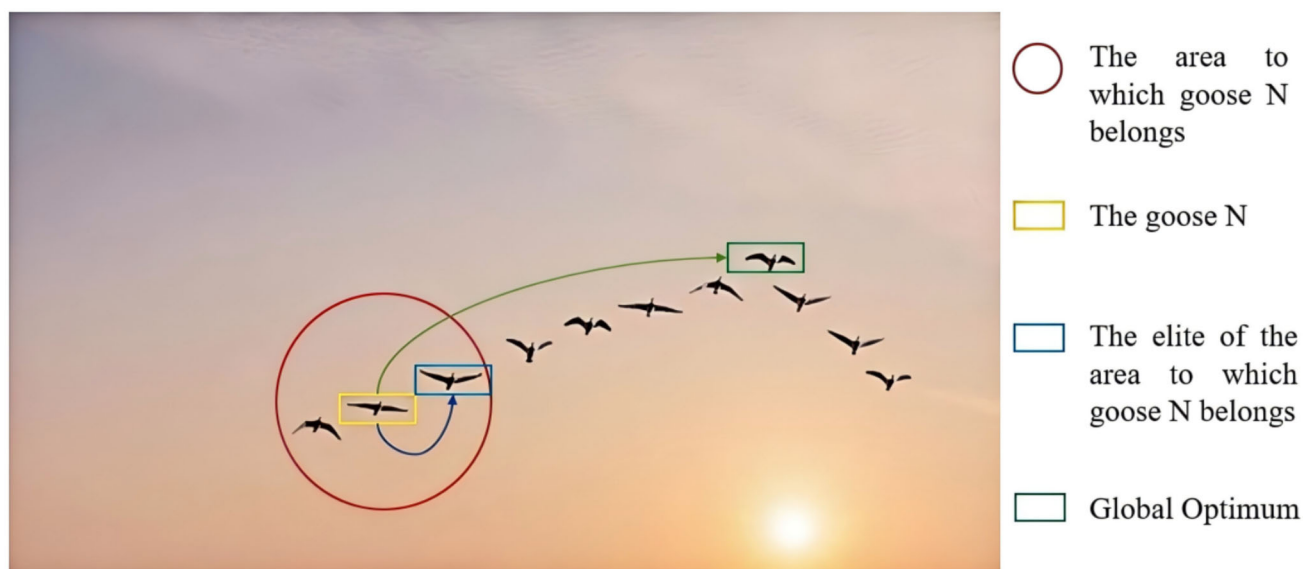


Fig. 5 As shown in the figure, the flying position of goose N will be affected not only by the geese in the global optimal position, but also by the geese in the regional optimal position in the area where it is located

Table 1 GHGBackbone lightweight hybrid backbone structure and parameter configuration

Layer no	Module	Input size	Output Ch	Stride/down	Repeats
0	ARHGStem	$H \times W \times 3$	48	$\times 4 (H/4)$	1
1	HGBlock	$H/4 \times W/4 \times 48$	128	1	4
2	DWConv	$H/4 \times W/4 \times 128$	96	$\times 2 (H/8)$	1
3	HGBlock	$H/8 \times W/8 \times 96$	512	1	4
4	DWConv	$H/8 \times W/8 \times 512$	1024	$\times 2 (H/16)$	1
5	GHGBlock	$H/16 \times W/16 \times 1024$	1024	1	4
6	GHGBlock	$H/16 \times W/16 \times 1024$	1024	1	4
7	GHGBlock	$H/16 \times W/16 \times 1024$	1024	1	4
8	DWConv	$H/16 \times W/16 \times 1024$	2048	$\times 2 (H/32)$	1
9	GHGBlock	$H/32 \times W/32 \times 2048$	2048	1	4

strategy and proposed a streamlined feature fusion strategy called SFF, which only retains mid- and high-level features for fusion. Experiments demonstrate that this simplified strategy yields superior performance in small object detection, while also reducing redundant computations. Furthermore, to enhance training efficiency and runtime performance, we introduce a novel hyperparameter tuning strategy based on an improved particle swarm optimization algorithm. This method shifts part of the optimization burden from the inference phase to the training process, thereby alleviating the additional inference cost typically associated with performance improvements. Extensive experiments on the VisDrone2019 test set show that GSR-Net achieves 93.9 FPS on an NVIDIA RTX 4090D GPU, outperforming RT-DETR by 35.8 FPS. In terms of detection accuracy, GSR-Net achieves a 0.5% gain in AP and a 0.3% gain in AP50, while reducing 19.6M parameters and requiring only 50.7 GFLOPs

for inference. The main contributions of this work are summarized as follows:

- (1) Design a lightweight and efficient backbone architecture, termed GHG, by combining HGNetv2 and Ghost convolution, which significantly reduces model complexity while maintaining detection performance.
- (2) Propose the SFF strategy that retains only mid-to-high-level features, enabling effective small object detection and improving inference speed in remote sensing imagery.
- (3) Introduce an improved particle swarm optimization-based hyperparameter tuning method during training, enhancing accuracy while reducing the computational load during inference.
- (4) Validate GSR-Net in real-world UAV aerial sensing scenarios, demonstrating its superior robustness against missed detections under high-speed flight conditions.

Table 2 Comparison of the current object detection models on the VisDrone2019 test dataset

Model	Input	Backbone	#Params (M)	GFLOPs	FPS _{bs=1}	AP	AP50	P	R	F1
<i>One-stage object detectors</i>										
TOOD-R50 [41]	(768, 1344)	R50	32.03	199	43.7	23.1	37.6	47.4	39.1	0.429
RetinaNet-R50 [23]	(768, 1344)	R50	36.15	210	56.7	19.6	32.3	42.9	33.9	0.379
MADNet-R50 [40]	(640, 640)	R50	33.57	74.7	73.1	23.3	37.1	48.2	37.9	0.424
ATSS-R50 [42]	(768, 1344)	R50	38.91	110	52.4	23.2	39.1	48.5	39.0	0.432
RTMDet-M [43]	(640, 640)	–	24.66	39.1	50.9	24.0	38.1	48.1	36.1	0.412
YOLOv8-M [24]	(640, 640)	–	25.84	78.7	135.5	20.6	35.3	49.4	36.5	0.420
YOLOv9-M [25]	(640, 640)	–	20.07	77.1	103.4	23.1	38.5	51.9	38.9	0.445
YOLOv10-M [26]	(640, 640)	–	16.46	63.5	105.4	21.0	35.9	49.1	37.0	0.422
YOLO11-M [27]	(640, 640)	–	20.03	67.7	117.1	21.8	37.3	50.4	38.3	0.435
YOLOv12-M [28]	(640, 640)	–	20.11	67.2	106.3	21.2	36.8	50.1	38.2	0.433
<i>Two-stage object detectors</i>										
Faster-RCNN-R50 [29]	(768, 1344)	R50	41.39	208	53.3	22.7	36.4	46.5	38.4	0.421
Cascade-RCNN-R50 [44]	(768, 1344)	R50	69.18	236	44.1	22.9	37.0	47.1	37.9	0.420
EfficientNet-b3 [45]	(512, 896)	R50	18.52	62.5	61.7	16.9	28.3	37.6	30.5	0.337
<i>DETR-based object detectors</i>										
DETR [31]	(768, 1344)	–	41.57	96.5	80.7	13.0	26.6	34.3	29.4	0.317
Deformable-DETR [51]	(768, 1344)	–	40.10	193	36.9	16.6	30.5	38.3	33.4	0.357
DAB-DETR-R50 [48]	(768, 1344)	R50	43.70	102	66.1	17.0	32.8	43.8	33.7	0.381
Conditional-DETR-R50 [49]	(768, 1344)	R50	43.45	101	73.4	16.1	31.6	40.8	32.9	0.364
DINO-R50 [50]	(768, 1344)	R50	47.55	274	27.5	24.7	43.1	52.0	43.7	0.475
RT-DETR-R18 [32]	(640, 640)	R18	19.88	57	78.1	21.3	37.4	55.3	40.1	0.465
RT-DETR-R50 (Baseline) [32]	(640, 640)	R50	41.97	129	58.1	22.4	39.0	57.7	40.3	0.475
<i>Our end-to-end object detector</i>										
GSR-Net (ours)	(640, 640)	GHG	22.37	50.7	93.9	22.9	39.3	58.0	40.5	0.477

For metrics like mAP, precision, recall and F1 score, higher values are better; for computational costs (parameters, GFLOPs) and inference time, lower values are preferred

The model shows better stability and practicality compared to existing methods, offering a feasible solution for UAV-based inspection and monitoring applications.

2 Related works

Recent advances in object detection have primarily focused on enhancing both detection accuracy and inference efficiency. Traditional CNN-based detectors such as RetinaNet [23], YOLO series [24–28], and MADNet[40] have demonstrated strong performance on general-purpose datasets, but often struggle with small object detection and real-time inference in aerial imagery. To address these issues, one-stage detectors have been explored. TOOD [41] introduces a task-aligned learning strategy to better balance classification and localization, while ATSS [42] proposes an adaptive training sample selection mechanism to bridge the gap between anchor-based and anchor-free methods. Both

approaches improve detection accuracy but can still be computationally intensive for UAV applications.

Real-time detection has attracted significant attention, leading to the development of efficient architectures. RTMDet [43] designs lightweight yet high-performance detection networks capable of real-time inference without substantial accuracy loss. Multistage frameworks such as Cascade R-CNN [44] focus on improving high-quality detection by employing progressively increasing IoU thresholds in a cascade structure, enhancing the precision of bounding box regression and classification at the cost of increased computational complexity. Meanwhile, backbone optimization has become a key strategy for balancing accuracy and efficiency. EfficientNet [45] demonstrates that compound scaling of depth, width, and resolution can yield a highly efficient backbone for visual tasks, inspiring lightweight network designs for embedded UAV platforms.

Transformer-based detectors have recently emerged as powerful alternatives due to their ability to model global context [46, 47]. DAB-DETR [48] introduces dynamic anchor



Fig. 6 Detection examples of the proposed model and the baseline model on VisDrone

Table 3 Results of ablation experiments

Configuration	GHG	SFF	R-PSO	#Params (M)	GFLOPs	FPS _{bs=1}	AP	AP50	P	R	F1
1. Baseline				41.97	129	58.1	22.4	39.0	57.7	40.3	0.475
2	✓			30.60	99.2	88.1	22.4	38.9	57.7	40.3	0.475
3		✓		34.54	82.0	72.1	22.6	39.1	57.8	40.4	0.476
4	✓	✓		22.37	50.7	93.9	22.7	39.1	57.9	40.4	0.476
Ours	✓	✓	✓	22.37	50.7	93.9	22.9	39.3	58.0	40.5	0.477

The ✓ symbol indicates an improvement in the structure of the corresponding ordinate

Table 4 Results of backbone ablation experiments

Methods	#Params (M)	GFLOPs	FPS _{bs=1}	AP	AP50
Baseline (R50)	41.97	129	58.1	22.4	39.0
MobileNetV4	39.50	11.32	72.3	19.1	33.7
EfficientViT	10.71	27.3	41.2	19.3	33.7
RepViT	13.31	36.4	69.4	20.8	36.5
Cswin Transformer	30.49	89.9	25.5	22.2	38.3
HGNetv2	32.00	108	72.1	22.4	38.7
GHGNet (Ours)	30.60	99.2	88.1	22.4	38.9

Table 5 Results of feature fusion ablation experiments

Methods	#Params (M)	GFLOPs	FPS _{bs=1}	AP	AP50	P	R	F1
Baseline (P3+P4+P5)	41.97	129	58.1	22.4	39.0	57.7	40.3	0.474
P3+P5	34.41	81.9	70.9	21.2	36.9	52.7	38.9	0.447
P4+P5 (SFF)	34.54	82.0	72.1	22.6	39.1	57.8	40.4	0.475

boxes as queries, improving query initialization and convergence speed. Conditional-DETR [49] employs conditional cross-attention to accelerate training and enhance localization accuracy. Building upon these advances, DINO [50] further incorporates denoising anchor boxes and hybrid query strategies to achieve both fast convergence and high detection performance. These Transformer-based methods provide strong motivation for adopting end-to-end architectures in UAV-based small object detection, as they naturally handle sparse object distributions and large scale variations inherent in aerial imagery. Moreover, Deformable-DETR [51] addresses the slow convergence and small object detection limitations of DETR by introducing deformable attention and multi-scale feature fusion, which significantly reduces computational cost while preserving fine-grained spatial details. These Transformer-based methods provide strong motivation for adopting end-to-end architectures in UAV-based small object detection, as they naturally handle sparse object distributions and large scale variations inherent in aerial imagery [52–56].

3 Methods

3.1 GHGBlock backbone

As a traditional classification backbone, ResNet-50 gradually reduces spatial resolution through successive strided bottleneck blocks. While this layer-by-layer compression facilitates global semantic abstraction, it dilutes fine-grained spatial information, hindering the localization and recall of small objects. Recently, the novel HGNetv2 backbone network, based on repeated feature fusion and multi-path information flow at the same resolution, has achieved impressive results in various network performance tests. We have

Table 6 Search space boundaries and description

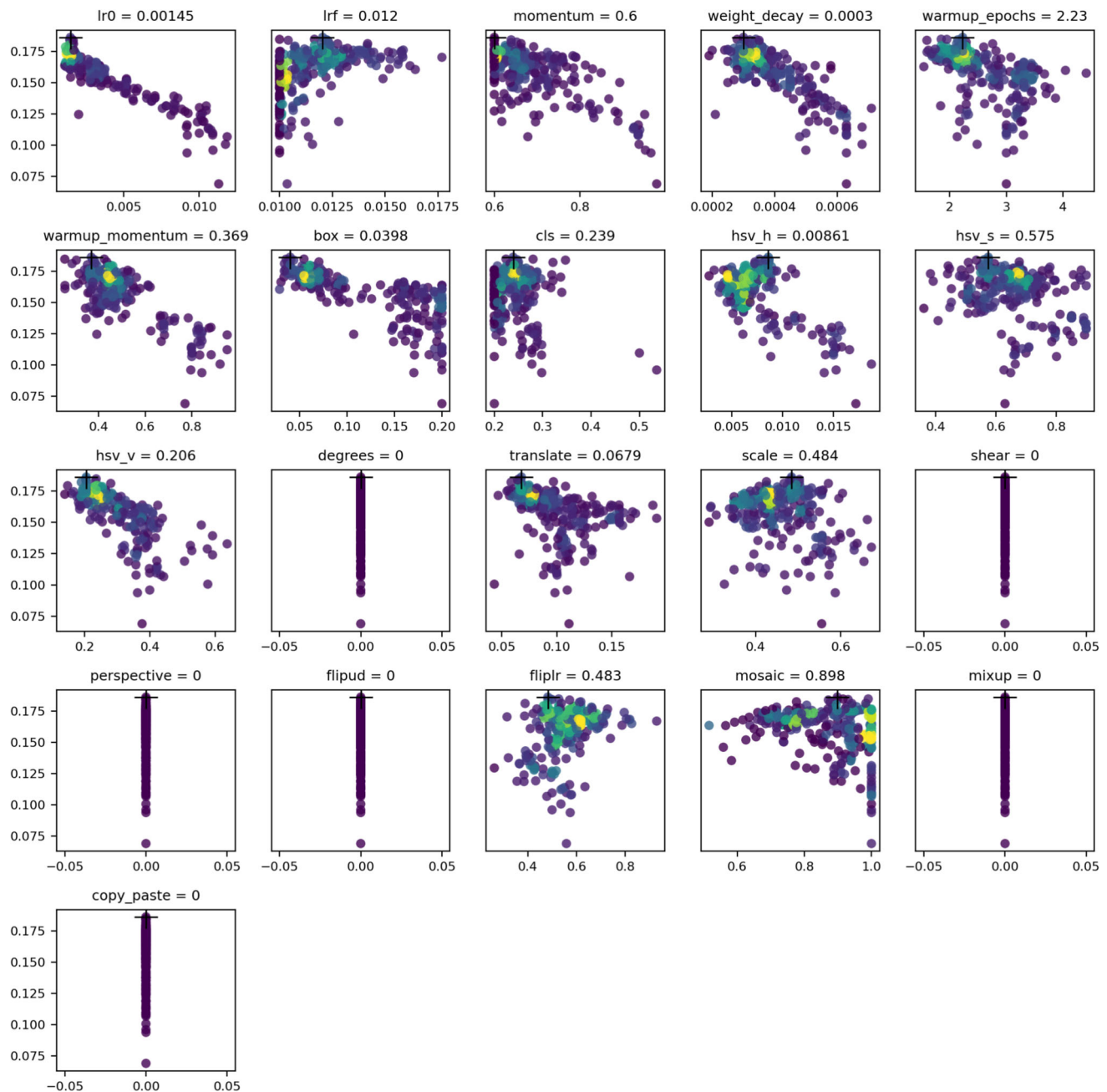
Parameter	Value range	Parameter	Value range
lr0	(1e-5, 1e-1)	degrees	(0.0, 45.0)
momentum	(0.6, 0.98)	scale	(0.0, 0.9)
weight_decay	(0.0, 0.001)	shear	(0.0, 10.0)
warmup_epochs	(0.0, 5.0)	perspective	(0.0, 0.001)
warmup_momentum	(0.0, 0.95)	flipud	(0.0, 1.0)
box	(0.02, 0.2)	fliplr	(0.0, 1.0)
cls	(0.2, 4.0)	mosaic	(0.0, 1.0)
dfl	(0.4, 6.0)	mixup	(0.0, 1.0)
hsv_h	(0.0, 0.1)	copy_paste	(0.0, 1.0)
hsv_s	(0.0, 0.9)		
hsv_v	(0.0, 0.9)		

improved upon the design of HGNetv2 and proposed GHG-Backbone. The GHGBackbone backbone retains the advantages of HGNetv2's repeated blocks at the same scale and multi-path fusion, while also incorporating key optimizations to significantly reduce computational overhead while maintaining high-order feature dimensionality.

First, a compact ARHGStem architecture was designed, as shown in Fig. 1. ARHGStem, located at the head of the backbone network, is responsible for efficiently extracting multi-scale shallow features at the input stage. Building on the HGStem architecture, ARHGStem introduces adaptive resizable convolution (ARConv). ARConv not only learns sampling position offsets based on input features but also dynamically adjusts the height, width, and number of sampling points of the convolution kernel to accommodate objects with large scale and shape variations in remote sensing imagery. The structural differences between ARConv and traditional convolution are shown in Fig. 2. Furthermore,

Table 7 Optimization results of the GSR-Net model based on the R-PSO

Methods	#Params (M)	GFLOPs	FPS _{bs=1}	AP ^{Test}	AP ^{Test} ₅₀
Baseline	22.37	50.7	93.9	22.9	39.3
GA	22.37	50.7	93.9	23.7	40.7
PSO	22.37	50.7	93.9	23.7	40.7
R-PSO (Ours)	22.37	50.7	93.9	24.0	41.2

**Fig. 7** R-PSO algorithm hyperparameter tuning scatter plot

ghost convolution is introduced into the backbone network to form an efficient feature extraction module, GHGBlock. The GHGBlock architecture is shown in Fig. 3. GHGBlock

first generates a small number of intrinsic channels and then expands them into a large number of ghost channels through low-cost deep operations, achieving high-dimensional out-

put with low FLOPs. Overall, GHGBackbone is a repetitive feature fusion architecture that introduces lightweight convolution kernels and convolution kernels customized for remote sensing detection. This enables it to achieve better small target detection performance per unit computational budget in remote sensing target detection tasks compared to traditional backbones such as ResNet-50.

3.2 SFF

To reduce computational redundancy and alleviate feature overlap, we designed the SFF module that involves only bidirectional interaction between P4 and P5. Specifically, the high-resolution P4 feature map is fused with the more semantically robust P5 feature map in a top-down and bottom-up manner: First, the P5 is upsampled and merged with the P4 to enhance the spatial detail representation of high-level semantic features. Subsequently, the refined P4 is downsampled and reintegrated with the P5 to supplement the global semantic representation with fine-grained information. Unlike traditional multilayer fusion, the P3 layer is intentionally discarded. On the one hand, P3 features primarily contain low-level texture information. While high in resolution, they contribute little to detection tasks and are prone to feature redundancy. On the other hand, preserving the complementary relationship between P4 and P5 allows for a balanced representation of both spatial detail and semantic information. Importantly, in remote sensing scenarios, many targets are small-scale objects, and the scale differences among similar targets are typically small. Complex backgrounds introduce significant noise into the P3 layer, weakening detection robustness. Therefore, we only retain the bidirectional fusion of P4 and P5 to extract more discriminative multi-scale features. By reducing the number of fusion levels, the proposed strategy effectively reduces the number of fusion operations and memory consumption, while achieving a good trade-off between efficiency and accuracy while maintaining sufficient multi-scale context information. The structures of SFF and the proposed model GSR-Net are shown in Fig. 4.

3.3 R-PSO for hyperparameter optimization

Hyperparameter optimization during the training phase is a key step in improving model detection accuracy without increasing the computational burden of the model. Inspired by the V-shaped migration behavior of wild geese (as shown in Fig. 5), R-PSO is proposed. In the flight of a goose flock, each goose (analogous to a particle) is influenced not only by the leading goose (global best) but also by the elite geese in the nearby region (regional best), facilitating collaborative optimization.

The hyperparameter search space is divided into K dynamically adjustable subregions (marked as region Q_k with red circles in Fig. 5). Within each region, particles are classified based on their fitness values Fit_i :

$$\mathbf{x}_i \in Q_k \quad \text{if} \quad \frac{k-1}{K} \max(Fit) < Fit_i \leq \frac{k}{K} \max(Fit) \quad (1)$$

The maximum fitness value of the current population, $\max(Fit)$, is used, and the boundaries of the regions are adaptively adjusted according to the particle distribution density to ensure that resources are tilted toward high-potential regions.

The velocity $\mathbf{v}_i$ update of each particle is driven by three components (the green arrows in Fig. 5 represent the global best, and the blue boxes represent the regional elites).

$$\begin{aligned} \mathbf{v}_i^{(t+1)} = & \omega \mathbf{v}_i^{(t)} + c_1 \mathbf{R}_1 \odot (\mathbf{p}_i^{\text{best}} - \mathbf{x}_i^{(t)}) \\ & + c_2 \mathbf{R}_2 \odot (\mathbf{g}^{\text{best}} - \mathbf{x}_i^{(t)}) \\ & + c_3 \mathbf{R}_3 \odot (\mathbf{r}_k^{\text{best}} - \mathbf{x}_i^{(t)}) \end{aligned} \quad (2)$$

$\mathbf{v}_i^{(t)}$ and $\mathbf{v}_i^{(t+1)}$ represent the velocity vectors of particle i in generations t and $t+1$, determining the particle's next movement direction and distance.

The inertia weight ω controls the influence of the previous velocity on the current velocity. A larger ω (e.g., close to 1) enhances global exploration capability, while a smaller ω (e.g., close to 0) promotes local exploitation. It is typically adjusted dynamically to balance the search process. The inertia weight δ decays nonlinearly with the number of iterations to balance exploration and exploitation:

$$\omega^{(t)} = 0.9 - 0.5 \cdot \left(\frac{t}{T_{\max}} \right)^{0.5} \quad (3)$$

where $T_{\max}$ represents the maximum number of iterations.

c_1 , c_2 , and c_3 are acceleration constants, while $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}_3$ are random matrices with elements uniformly distributed in $[0, 1]$. Through element-wise multiplication ($\odot$), these random matrices introduce stochastic perturbations, ensuring diversity in particle updates and preventing premature convergence.

$\mathbf{p}_i^{\text{best}}$: The particle's own search experience (the "Goose N" marked by the yellow box in Fig. 5).

$\mathbf{g}^{\text{best}}$: The best-performing hyperparameter configuration in the population (the "Global Best" indicated by the green arrow in Fig. 5).

$\mathbf{r}_k^{\text{best}}$: The particles with the highest fitness within the sub-region (the "Regional Elite" marked by the blue box in Fig. 5).

A fitness function is defined that balances accuracy, efficiency, and geometric consistency:

$$\text{Fit} = 0.9 \cdot AP + 0.1 \cdot AP_{50} \quad (4)$$

where AP and AP_{50} are model accuracy indicators, and the specific calculation methods will be given in the experimental part.

The particle with the highest fitness in each subregion Q_k is selected as the regional elite (as shown by the blue box in Fig. 5).

$$\mathbf{r}_k^{\text{best}} = \underset{x_j \in Q_k}{\operatorname{argmax}} \text{Fit}_j \quad (5)$$

4 Results

4.1 Datasets

First, we verified the model performance on the authoritative UAV small target dataset VisDrone2019 [44]. Secondly, we built the Shihezi University campus aerial photography dataset SUVD and conducted drone inspection experiments in real environments, demonstrating the advancement and practicality of our model in diverse aerial photography scenarios.

The VisDrone2019 dataset is an authoritative resource in the field of international drone vision. It contains diverse multi-scene and multi-task images taken by various types of drones in 14 cities in China in urban and rural environments, sparse and dense scenes, cloudy and sunny weather, and day and night lighting conditions. The dataset contains 10 types of targets, including pedestrians, people, bicycles, cars, vans, trucks, tricycles, awning tricycles, buses, and motorcycles. According to the official partitioning scheme of the VisDrone2019 challenge, we use a pre-split dataset: the training set contains 6471 images, the validation set contains 548 images, and the test set contains 1610 images.

The SUVD dataset is a collection of aerial photography taken at Shihezi University within a 120 m height limit. In dataset construction, similar to traditional methods, the images were cropped to 1024×1024 pixels, preserving their original aspect ratio and maintaining a 200-pixel overlap during segmentation. After filtering and removing unlabeled images after segmentation, the dataset contains 2,480 training images and 1,062 validation images. Experiments on the SUVD dataset validate the model's performance on the validation set and its practicality in real-world aerial photography activities.

Table 8 Performance comparison of different object detection models in a real-world scenario

Model	Size (Pixels)	FPS	AP	MR
YOLOv10-M	960×540	15.9	46.9	0.07
YOLO11-M	960×540	19.2	48.3	0.06
YOLOv12-M	960×540	16.0	46.9	0.04
RT-DETR-R50	960×540	14.7	49.0	0.04
GSR-Net	960×540	17.3	49.8	0.01

4.2 Evaluation metrics

In this section, we use several key indicators to comprehensively evaluate the performance of the model on different datasets, covering aspects such as accuracy, robustness, and model complexity. First, as shown in formula (6), precision is used to measure the proportion of true positives in all predicted bounding boxes, reflecting the model's ability to avoid false positives in the detection results; recall, as defined in formula (7), represents the proportion of correctly detected objects among all real objects, reflecting the model's missed detection. In order to strike a balance between precision and recall, we further introduce the F1 score (formula (8)), which is the harmonic mean of the two and can more comprehensively evaluate the overall detection performance of the model in different scenarios.

$$\text{Precision} = P = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = R = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = \frac{2 \times P \times R}{P + R} \quad (8)$$

Among the core indicators of object detection tasks, mean average precision (mAP) plays an important role. Among them, mAP50 (abbreviated as AP50) measures the detection quality by calculating the average precision of the predicted box when the intersection over union (IoU) threshold is 0.5, as shown in formula (4). Its value can intuitively reflect the detection level of the model under normal conditions. mAP50:95 (abbreviated as AP) is further averaged within the IoU threshold range of 0.5 to 0.95 (with a step size of 0.05), as shown in formula (5). This indicator is more rigorous and can comprehensively characterize the detection performance of the model under different matching criteria. Therefore, it is widely used in current mainstream research.

$$AP_{50} = \frac{1}{N} \sum_{i=1}^N AP_i (\text{IoU} = 0.5) \quad (9)$$

$$AP = \frac{1}{10N} \sum_{t=0}^9 \sum_{i=1}^N AP_i(\text{IoU} = 0.5 + 0.05t) \quad (10)$$

In addition to detection accuracy, we also introduce several indicators related to model complexity and inference efficiency to evaluate the potential of the model for practical deployment. First, frame rate (FPS) measures the number of image frames the model can process per second. It is a critical criterion for real-time detection systems, and generally, models need to achieve 40–50 FPS or higher to meet the real-time requirements of industrial inspection, drone monitoring, and autonomous driving. Second, floating-point operations (FLOPs) reflect the computational workload of the network during the inference phase. A lower FLOPs value indicates lower computational complexity and higher inference efficiency. Third, the number of parameters (Params), usually measured in millions (M), represents the model size and is an important reference for lightweight network design. Models with fewer parameters not only reduce storage and transmission costs but also exhibit better adaptability for deployment on resource-constrained edge devices.

4.3 Implementation details

The proposed model was implemented on a workstation equipped with an Intel Core i5-13600KF CPU and an NVIDIA GeForce RTX 4090D GPU (24 GB), running Ubuntu 22.04. All experiments were conducted using PyTorch 2.1.0, CUDA 12.1, and Python 3.10.0. The deep learning models were developed based on the open-source frameworks MMDetection, Ultralytics, and PyTorch.

For training, the model was initialized from scratch without any pretrained weights. The AdamW optimizer was employed with an initial learning rate of 1×10^{-4} , momentum of 0.9, and weight decay of 1×10^{-4} . To enhance the model's robustness, Mosaic data augmentation was applied. The maximum number of training epochs was set to 300. To ensure fairness in comparison across different configurations, the batch size was consistently set to 6 for all experiments (Table 1).

4.4 Comparative result analysis of different detection models

To validate the effectiveness of the proposed model, GSR-Net was compared with existing state-of-the-art object detection methods on the VisDrone2019 benchmark. The specific comparison algorithms and experimental results are shown in Table 2. In terms of detection accuracy, DINO-R50 achieved the highest overall AP (24.7) and AP₅₀ (43.1) among all compared methods, demonstrating strong detection capabilities, but at a very high computational cost (274 GFLOPs). In

contrast, our proposed GSR-Net achieved an AP of 22.9 and an AP₅₀ of 39.3, outperforming YOLOv8-M (AP=20.6), YOLOv10-M (AP=21.0), and YOLOv12-M (AP=21.2). It also achieved the highest F1 score (0.477), demonstrating the best balance between precision and recall. GSR-Net also achieved a recall of 40.5%, surpassing most single-stage detectors (e.g., ATSS-R50: 39.0) and the DETR baseline (RT-DETR-R50: 40.3). Its precision (58.0) was the highest among all models, indicating more reliable predictions and lower false positive rates.

In terms of efficiency and model complexity, GSR-Net requires only 22.37M parameters and 50.7 GFLOPs, significantly lower than DINO-R50 (47.55M parameters, 274 GFLOPs) and Cascade-RCNN-R50 (69.18M parameters, 236 GFLOPs). In terms of inference speed, GSR-Net achieved 93.9 FPS, surpassing most two-stage and DETR baseline methods and comparable to lightweight YOLO models (e.g., YOLOv9-M: 103.4 FPS). While some models (e.g., DINO-R50) achieve slightly higher AP, their high computational cost and slow inference speed make them unsuitable for real-time applications. In contrast, our proposed GSR-Net achieves the best balance between accuracy, efficiency, and computational complexity, achieving the best overall performance. Some detection examples of the proposed model and the baseline model on VisDrone are shown in Fig. 6.

4.5 Ablation experiments

Table 3 shows the results of ablation experiments, illustrating the contributions of the different components of our proposed model. The baseline model without any improvements has 41.97M parameters and 129 GFLOPs, achieving 58.1 FPS, an AP of 22.4, an AP₅₀ of 39.0, and an F1 score of 0.475.

Adding GHGBlock (configuration 2) improves inference speed to 88.1 FPS, reduces parameter size to 30.60M, and maintains an overall AP similar to the baseline (22.4), indicating that GHGBlock's primary contribution is efficiency. Introducing SFF (configuration 3) reduces model complexity to 34.54M parameters and 82.0 GFLOPs, while also improving inference speed to 72.1 FPS. Furthermore, SFF slightly improves AP to 22.6 and F1 score to 0.476, demonstrating that the P4.5 layer can enhance detection performance while reducing computational cost. When GHGBlock and SFF are added simultaneously (configuration 4), the model achieves a better balance, with 22.37M parameters, 50.7 GFLOPs, 93.9 FPS, 22.7 AP, and 0.476 F1 score, improving both efficiency and detection performance. Finally, incorporating R-PSO hyperparameter tuning into the full model further improves AP to 22.9 and AP₅₀ to 39.3, achieving slight improvements in precision, recall, and F1 score, while maintaining its lightweight and fast inference properties. Overall, these ablation studies verify that each proposed component

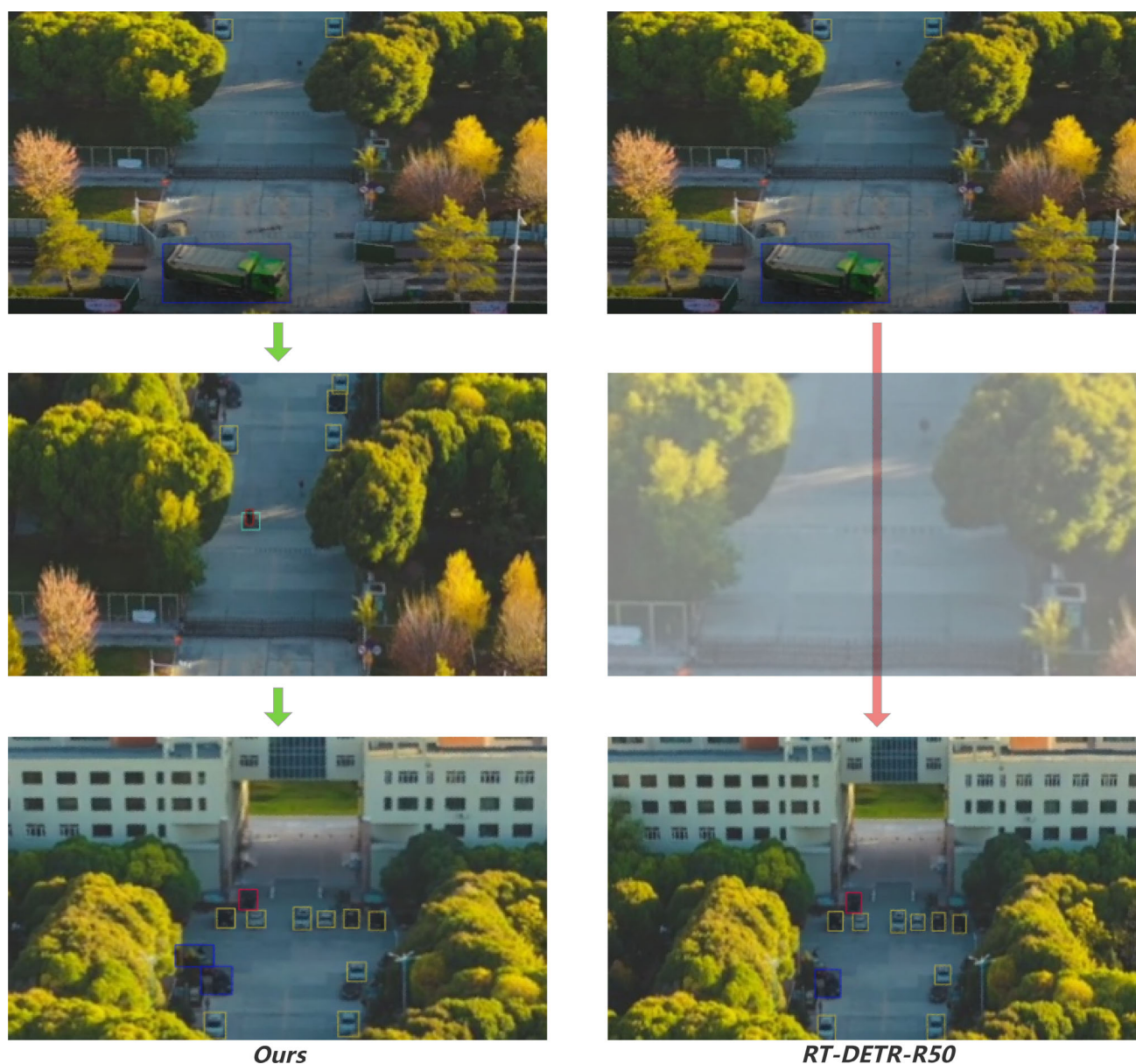


Fig. 8 Real-world low-altitude aerial photography detection results

makes a positive contribution, and their combination in the full model achieves an optimal trade-off between accuracy, efficiency, and model complexity.

4.5.1 Ablation analysis for backbone

Table 4 presents the ablation study on different backbone networks. The baseline model using ResNet-50 has 41.97M parameters and 129 GFLOPs, achieving 58.1 FPS with an AP of 22.4 and AP₅₀ of 39.0.

Replacing the backbone with lightweight networks such as MobileNetV4 or EfficientViT significantly reduces the model complexity and computational cost. Specifically,

MobileNetV4 achieves only 19.1 AP and 33.7 AP₅₀ with 72.3 FPS, while EfficientViT reaches 19.3 AP and 33.7 AP₅₀ at 41.2 FPS, indicating a drop in detection accuracy despite improved or reduced inference speed. RepViT improves AP to 20.8 and AP₅₀ to 36.5 while maintaining a high FPS of 69.4. CSWin Transformer and HGNetv2 achieve better accuracy (AP=22.2–22.4, AP₅₀=38.3–38.7) but differ in efficiency, with CSWin Transformer being slower (25.5 FPS) and HGNetv2 faster (72.1 FPS). Our proposed GHGNet backbone achieves a competitive AP of 22.4 and AP₅₀ of 38.9 while providing the highest inference speed of 88.1 FPS among high-accuracy models. These results demonstrate that GHGNet effectively balances detection accuracy,

model complexity, and inference efficiency, validating its suitability as a lightweight yet high-performance backbone.

4.5.2 Ablation analysis for SFF

Table 8 presents the results of the ablation experiments comparing different feature fusion strategies. Using P3+P4+P5 as the baseline, the model has 41.97M parameters, 129 GFLOPs, and achieves 58.1 FPS, with an AP of 22.4, AP₅₀ of 39.0, and an F1 score of 0.474.

When removing the P4 layer and using only P3+P5, both the number of parameters and GFLOPs decrease to 34.41M and 81.9, respectively, while the inference speed increases to 70.9 FPS. However, the AP drops to 21.2, AP₅₀ to 36.9, and F1 score to 0.447, indicating that the absence of the middle-level feature P4 negatively affects detection performance, especially for small- and medium-sized objects.

By adopting P4+P5 (SFF), the model maintains a similar parameter size (34.54M) and GFLOPs (82.0) while further improving the inference speed to 72.1 FPS. Meanwhile, AP and AP₅₀ increase to 22.6 and 39.1, with an F1 score of 0.475, surpassing both the baseline and the P3+P5 combination. These results demonstrate that removing the lowest-level P3 layer effectively reduces model complexity and improves efficiency, while retaining the P4 layer preserves or slightly enhances detection accuracy, particularly for medium-sized objects. Overall, the P4+P5 feature fusion strategy achieves the best balance between accuracy and efficiency, validating the design of our proposed model (Table 5).

4.6 Hyperparameter tuning experiment results

All hyperparameter tuning experiments were conducted using a unified hardware configuration to ensure comparable results. During the hyperparameter tuning phase, 22 hyperparameters with significant impact on model performance were selected. The detailed search ranges for these parameters are listed in Table 6. Each algorithm iteratively searches for the optimal hyperparameter combination within a predefined search space. Specifically, in each iteration, the hyperparameter tuning algorithm substitutes candidate hyperparameter combinations into the model training process and evaluates performance on the validation set.

During hyperparameter optimization, all algorithms used a population size of 30, with a total of 30 iterations. 30 hyperparameter combinations were evaluated per generation, and each combination was trained for 10 generations to identify performance trends. The entire search process involved 9,000 model training cycles. To save time, we randomly selected one-tenth of the original VisDrone2019 dataset as the training set for hyperparameter tuning. A comparison of the optimization results of different methods is reported in Table 7.

From Table 7, it can be observed that all optimization strategies achieve significant improvements over the Baseline configuration. Specifically, both GA and standard PSO yield a performance gain of **+0.8% AP** and **+1.4% AP₅₀** compared to the manually tuned baseline, demonstrating the effectiveness of systematic hyperparameter tuning. Notably, our proposed **R-PSO** achieves the best overall performance, further improving the model's AP to **24.0%** and AP₅₀ to **41.2%**, which surpasses GA and PSO by **+0.3% AP** and **+0.5% AP₅₀**, respectively. Importantly, these gains are achieved without any changes in parameter size, FLOPs, or inference speed, confirming that the improvements stem purely from superior hyperparameter selection. The scatter plot recording the hyperparameter exploration during the iteration process is shown in Fig. 7.

The performance advantage of R-PSO mainly stems from the introduction of a regional guidance mechanism that optimizes the search process, enabling particles to perform more directional path exploration within key regions during global search. This approach maintains efficient coverage of the search space while improving search efficiency and reducing the risk of convergence to local optima. Compared with conventional PSO and GA methods, this strategy achieves a better balance between global exploration and local exploitation, allowing the optimization process to converge more stably toward better solutions in complex scheduling environments, thereby resulting in superior performance in overall operational efficiency and energy consumption optimization.

4.7 Real environment test

In this experiment, the flight platform was a DJI Air 2 S drone manufactured by Shenzhen DJI Innovations Technology Co., Ltd. (Shenzhen, Guangdong, China). Image inference and model deployment were conducted on the NVIDIA Jetson Orin Nano 8GB edge-computing platform, which provides powerful embedded computing capabilities and enables real-time detection tasks under resource-constrained conditions.

Table 8 summarizes the performance of different detection models in actual scenarios is summarized. It is particularly noteworthy that the models used in this experiment have all undergone hyperparameter tuning based on R-PSO. The FPS and AP values were obtained from training on the SUVD dataset, while the missed detection rate (MR) was measured during actual UAV flight tests (SUVD Video Collection). The results indicate that although RT-DETR-R50 achieves competitive detection accuracy (49.0 AP), its inference speed is limited to 14.7 FPS, restricting real-time deployment efficiency. In contrast, the lightweight YOLO11-M and GSR-Net achieve more balanced performance on the Jetson Nano. YOLO11-M reaches 19.2 FPS while maintaining 48.3 AP, significantly outperforming YOLOv10-M and YOLOv12-M. Meanwhile, GSR-Net achieves the best detec-

tion accuracy (49.8 AP) and the lowest MR (0.01), while maintaining a competitive inference speed of 17.3 FPS. The FPS drop from 93.9 on an RTX 4090 to 17.3 on the Jetson Orin is mainly due to the difference in CUDA core count—16,384 on the RTX 4090 versus 512 on the Jetson Orin. These results demonstrate that models designed with lightweight principles can effectively reduce performance degradation during embedded deployment, achieving an optimal trade-off between detection accuracy and inference efficiency.

Further analysis centered on the missed detection rate shows that higher detection accuracy does not necessarily lead to fewer missed detections. For instance, although YOLOv12-M (46.9% AP) performs slightly worse than RT-DETR-R50 (49.0 AP), both models exhibit the same MR of 0.04. This finding suggests that MR is influenced not only by accuracy but also by inference speed. As illustrated in Fig. 8, when detecting continuously moving objects, models with lower frame rates RT-DETR-R50 (14.7 FPS) are more prone to missing critical frames, thereby failing to capture transient targets. In contrast, GSR-Net maintains both high accuracy and a relatively fast inference rate (17.3 FPS), resulting in fewer missed detections and stronger robustness in dynamic aerial scenes. Consequently, GSR-Net demonstrates the best overall performance among the tested models.

5 Conclusions

In this paper, GSR-Net is proposed—a novel lightweight Transformer-based object detection model specifically tailored for real-time applications in UAV remote sensing. Addressing the critical demand for balancing high accuracy with rapid inference in aerial imagery, GSR-Net incorporates a series of architectural and methodological innovations.

The GHGBlock backbone was constructed by integrating the efficient HGNetv2 with Ghost convolution and an adaptive resizable convolution (ARConv). This design significantly reduces model parameters and computational overhead, ensuring suitability for deployment on resource-constrained platforms while maintaining strong feature extraction capabilities. Additionally, the conventional multi-scale fusion process was streamlined through the introduction of the SFF module, which focuses on bidirectional interaction between P4 and P5 features. This approach mitigates redundancy from lower-level layers, enhancing small object detection in complex remote sensing scenes and improving inference speed. To optimize training, the R-PSO algorithm was employed to automatically fine-tune hyperparameters, thereby offloading computational burden from the inference stage and ensuring a final model that is both efficient and accurate.

Extensive experiments on the VisDrone2019 dataset highlight the superior performance of GSR-Net. The model

achieves real-time inference at 93.9 FPS on an NVIDIA RTX 4090D GPU—outpacing RT-DETR by 35.8 FPS—while also delivering improved accuracy with an increase of 1.6% in AP and 2.2% in AP50. Importantly, these gains are achieved with 19.6M fewer parameters and 50.7 GFLOPs. Further validation through real-world drone tests on the custom SUVD dataset and during high-speed aerial photography confirms the model's robustness and practicality, demonstrating resilience against challenges such as frame loss and missed detections.

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Author Contributions Dear Editor, I have revised this paper according to your suggestions. Please review it.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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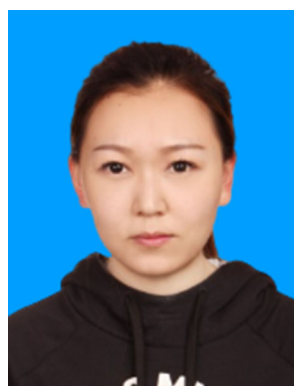
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